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Heart Disease Predictions using CDC Survey Data

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Abstract

Heart disease is a very serious ailment with many factors that lead to its development. The primary goal of this research project is to investigate the factors that contribute to heart disease and to develop a machine learning model that can aid in its prediction. The model could assist medical professionals to predict the likelihood of a patient developing heart disease by looking at contributing risk factors and common comorbidities. This research project makes use of various machine-learning approaches to identify significant factors and to create a model with the highest prediction accuracy. 2022 Survey data collected by the Centres for Disease Control and Prevention (CDC) was used to construct the models. The research project focused on F2 scores, Recall, to determine the model with the best fit. Ultimately, a Logistic Regression model with reduced dimensionality using Principal Component Analyses (PCA) was selected, the model could very accurately predict True Negatives but due to an imbalanced class problem, only 5,6% of sample entries had heart disease, the model could not accurately predict True Positives. While the model often predicted False Positives, this was seen as beneficial since it could be used as an indicator to medical professionals that the patient exhibited many high-risk factors that that contribute to developing heart disease and can thus be used as an early pre-emptive indicator that the patient has high risk of developing the condition.

Research Question

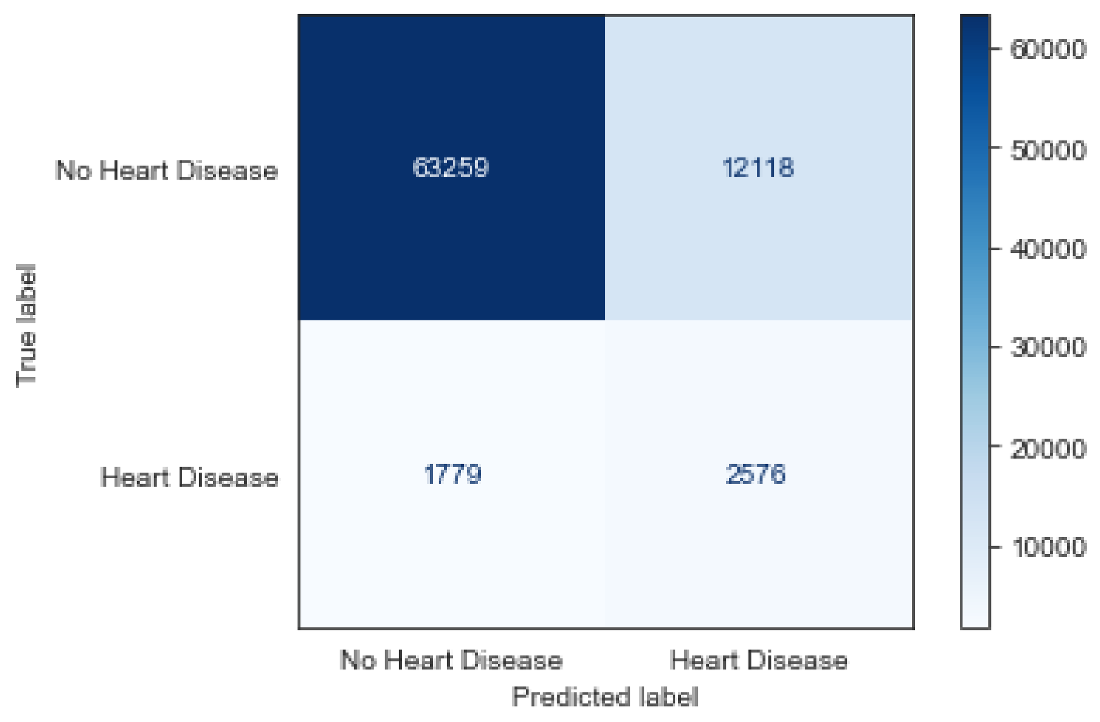
The primary question that this research project aims to answer is whether it is possible to construct a machine learning model that makes use of identified significant high-risk factors to accurately predict heart disease in patients before the ailment develops, allowing medical professionals to take preventive steps to assist the patient before the ailment becomes too serious.

Research Objectives

- To construct a data set containing medical information of patients from CDC survey data.
- To identify factors that are significantly related to the development of heart disease.
- To construct a model that is capable of predicting the likelihood of a patient developing heart disease.

Methodology

- 2022 Survey data collected on patients by the CDC was used in identifying significant factors that lead to heart disease. The dataset contained a large number of dimensions in the feature space (278 features excluding labels) and consisted of 401 958 samples.
- Initially, an intuition-based reduction was used to remove all irrelevant features (e.g. cell phone vs. landline).
- Due to responses in the survey being voluntary, a large number of samples had null entries. Samples with null entries were either deleted or imputed with feature means/medians where appropriate.
- Most features were categorical variables and were either ordinally encoded (e.g. GeneralHealth) or nominally encoded (e.g. Race) by making use of One Hot Encoding.
- A further feature reduction was performed by looking at a feature's correlation to heart disease with a cut-off set at 0.15. This reduced the feature space to 5 significant features; HeartDisease, GeneralHealth, AgeGroup, Stroke, COPD, and DifficultyWalking.
- Initially, the models performed very poorly due to machine learning algorithms being sensitive to the balance of training labels, undersampling was thus performed to improve class balance.
- Samples were shuffled and split with the ratio 80-20 for training and testing respectively with the training set being further split into a validation set.
- Principal Component Analyses (PCA), Logistic Regression, K-Nearest Neighbours (KNN), Random Forest, Naive Bayes Classifier, Multi-layer Perceptron Classifier, and Easy Ensemble Classifier algorithms were implemented.



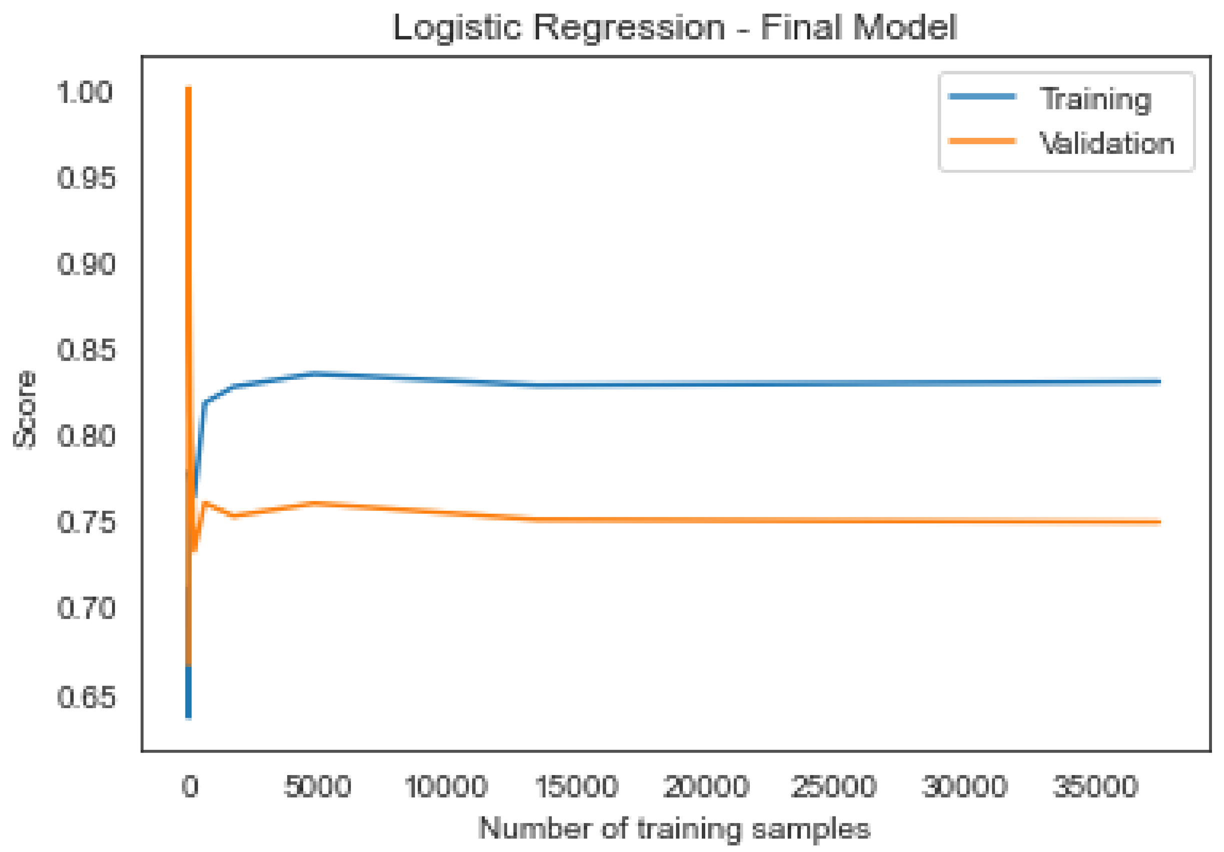
Results

Final Model:

The final Logistic Regression model with reduced dimensions can be seen below. An attempt was made to change the regularization parameter using grid search but no improvements were detected. The model performed the best in terms of F2 Scores and was able to very accurately predict True Negatives but due to an imbalanced class problem, the model could not accurately predict True Positives. While the model often predicted False Positives, this was seen as beneficial since it could be used as an indicator to medical professionals that the patient exhibited many high-risk factors that contributed to the development of heart disease and can thus be used as an early pre-emptive indicator that the patient is at high-risk of developing the condition.

Significant Features:

During feature exploration and selection, significant features that contribute to heart disease were identified and analysed with a patient's general health being the primary indicator. Patients with poor general health had a significantly higher likelihood of developing heart disease with already present chronic obstructive pulmonary disease (COPD) also playing a significant role. The likelihood was further increased if the patient has experienced a stroke in the past and had difficulty walking or moving. Finally, a patient's age also plays a role in the development of heart disease, the older the patient is, the more likely they are to develop heart disease.



Conclusion

Machine learning can effectively be used to assist medical professionals in identifying significant risk factors that lead to the development of heart disease as well as predicting which patients have a high risk of developing the condition. A Logistic Regression algorithm with reduced dimensionality using PCA displayed the best results when looking at F2 scores, Recall, when making predictions about heart disease. A class imbalance problem could affect the ability of algorithms to accurately predict True Positives, additional survey data of patients with heart disease is thus required to improve the model's accuracy and optimisation.